

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456048.67921 +/- 0.00039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.034
Transit depth (Rp/Rs)^2	3.79 +/- 1.34 [%]
Orbital Inclination (inc)	79.62 +/- 2.39
Airmass coefficient 1 (a1)	1.22 +/- 0.023
Airmass coefficient 2 (a2)	-0.126 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.87 %
Best Comparison Star	None
Optimal Aperture	3.76
Optimal Annulus	13.74
Transit Duration (day)	0.041 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.034 vs 0.1594 (deviation 1.02σ)
Duration fit vs published	0.041 d vs 0.0512 d (deviation 0.67σ)
Mid-time O–C	0.8 min
Depth SNR	5.6
Residual scatter	1.87 %

Light curve

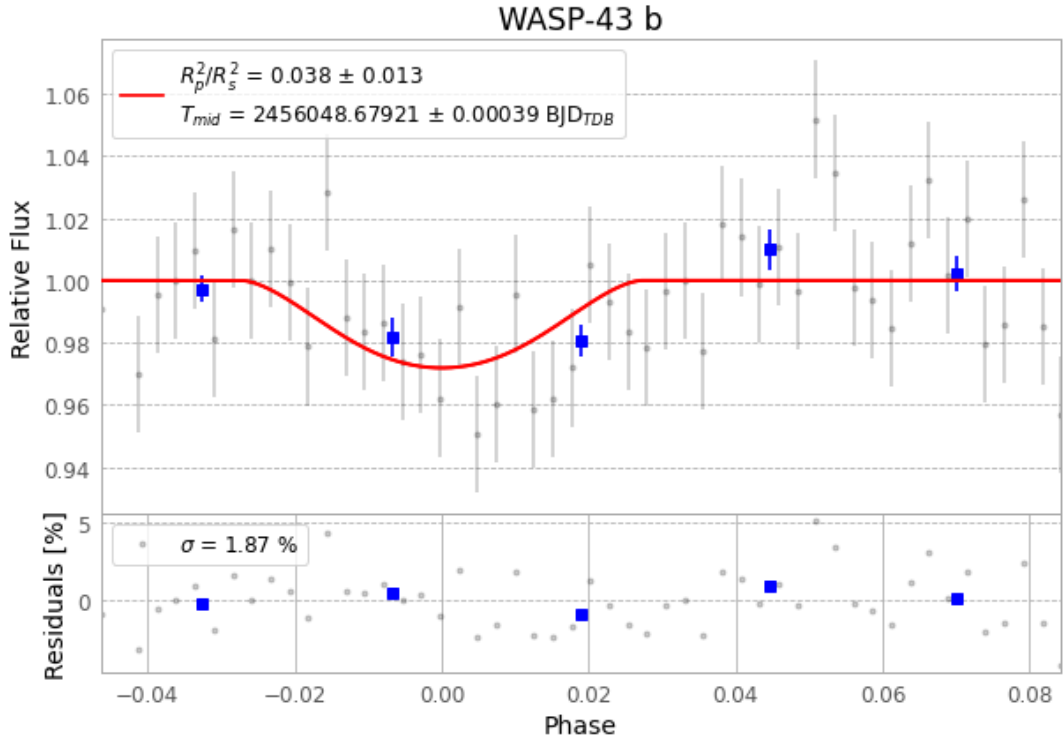


Figure 1 — FinalLightCurve_WASP-43 b_2012-04-30.png (SHA-256 aa492b757e3289e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e1ae29161f584496...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

52 FITS frames (34 MB), WASP-43120501031813.FITS → WASP-43120501055112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-120501.log (27030c46902a...), FinalLightCurve_WASP-43 b_2012-04-30.png (aa492b757e32...), FinalLightCurve_WASP-43 b_2012-04-30.csv (7d25874ee82d...)

Compute resources

Wall time 108.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:59Z.